

Matthew Segarra

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Lakeland, FL (Open to Relocate - No Assistance Required)

EDUCATION

Florida Polytechnic University, GPA: 4.00/4.00

Lakeland, FL

Bachelor of Science (BS): Mechanical Engineering – Advanced Topics

May 2026

Master of Science (MS): Mechanical Engineering

Graduating May 2027

Relevant Courses: Principles of Electrical Engineering, Computer Manufacturing and Control, Human Robotics

WORK EXPERIENCE

Junior Battle Bots Robotics Instructor

June 2026 – July 2026

iD Tech – Summer Tech Camps

Tampa, FL

- Instructed **40+** students (ages 6–13) in designing, building, & coding **VEX robots** for custom competitions.
- Taught core **Python** concepts & **Pygame** library, mentoring 4 students to **develop 5+ original 2D games**.
- Cultivated an engaging learning environment over **6** weekly sessions, reaching over 95% satisfaction.

Power Grid Operations (PGO) Distribution Intern

May 2025 – August 2025

Duke Energy

Largo, FL

- Designed an evolving power grid for **overhead & underground** coastal residential & commercial projects using **Design Lite**, **Argus GIS**, **Pole Foreman**, & **SEDS** improving the efficiency & reliability of the grid.
- Orchestrated **post-hurricane restoration** of Madeira Beach Elementary by removing temporary infrastructure.
- Contributed to upgrading the aging distribution infrastructure including leaning/broken poles & damaged cables to **improve service reliability** by over **20%** for over **15 direct customers**.

Community Director & Resident Assistant

August 2024 – Present

Florida Polytechnic University Housing

Lakeland, FL

- Mentored **4** Resident Assistants while acting as a resource for **110+** total residents.
- Organized monthly floor events & **\$1000 budget** community programs to foster a positive living environment.

PROJECTS

GYN Colpo-Pneumo Occluder | *Injection Molding, Material Science, R&D*

September 2025 – May 2026

- Manufactured custom **injection-molded silicone** Colpo-Pneumo Occluders with 40% barium sulfate formulation, introducing **radiopacity** and enhancing mechanical durability of devices at Lakeland Regional Health.
- Evaluated part integrity through **SOLIDWORKS** pressure simulations, **X-Rays**, & professional feedback.

Grip Aid Rehabilitation Device | *SOLIDWORKS, EMG Sensor, Arduino*

November 2025 – December 2025

- Developed a hand exoskeleton utilizing surface electromyography (**sEMG**) & a 40 kg·cm servo motor to provide motorized grip assistance upon a threshold-based trigger.
- Rapidly prototyped **3D-printed** structural components materials to validate **kinematic fit** & range of motion

Loader 5 Torsional Tester | *SOLIDWORKS, Arduino, Strength of Materials, 3D Printing*

August 2024 – May 2025

- Designed & manufactured a functional torsional tester using **Prusa MINI+** 3D-printed parts, steel bars, a DC motor, potentiometer, load cell, & Arduino Uno all under a budget of \$100 & available lab equipment.
- Tested the load & **angular displacement** for 20+ 3D printed PLA samples of varying infills at 100% success rate.

LEADERSHIP

VP of Professional Development | *Society of Hispanic Professional Engineers*

January 2026 – Present

- Delivered professional development workshops & networking events to connect members with recruiters.

Assistant Engineer | *Purple Fire Vex Robotics*

August 2023 – May 2024

TECHNICAL SKILLS

Certifications: FE, CSWP | **Software:** SOLIDWORKS, MATLAB, PLC, C, Python, Maximo, Pole Foreman, Design Lite, MS Office, Arduino IDE, Prusa Slicer, G-Code, RCA | **Hardware:** Arduino Uno R3, Prusa MINI+